Representing Content and Meaning in Discourse

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I. Introduction.

The goal of the presentation is:

to describe a new dynamic system in which we can explicitly represent both the static **contents** (propositions) and the dynamic **meanings** (context-change potentials) of sentences in discourse.

I. Introduction.

The central observations:

- modal subordination (e.g. *A wolf might come in. It would eat John*) provides a paradigm for 'extracting' static contents from dynamic meanings and determining relations between contents in discourse;

- natural language discourse treats contents on a par with individuals – so, we should represent discourse reference to contents (propositions) just as we represent discourse reference to individuals.

I. Introduction.

The system makes possible an integrated analysis of several phenomena:

- entailment relations established *within discourse* by particles like *therefore* / *hence* (e.g. *A man saw a woman. Therefore, he noticed her.*);
- modal subordination;
- the parallels between plural anaphora in the individual and modal domains.

I. The Structure of the Presentation.

I. Introduction: terminology and motivation.
II. Arguments for the explicit representation of contents in discourse.
III. Constraints on the representation of content and meaning in discourse.
IV. Dynamic Plural Logic with Contents (DPLC).
V. Conclusion.
VI. Appendix: Some consequences and extensions.

I. Meaning vs. Content.

The Partee marble examples:

- (1a) and (2a): different meanings (different anaphora licensing potential), but the same content.

(1) a. I dropped ten marbles and found all of them, except for one.

b. It is probably under the sofa.

(2) a. I dropped ten marbles and found only nine of them.

b. #It is probably under the sofa.

I. Content and Entailment.

Entailment is a relation between contents.

(1a) entails and is entailed by (2a),

i.e. they express the same content, the same *factual information* (the same information about the world – they have the same truth-conditions).

Meaning is more than factual information: it is also (among other things) *discourse information,* e.g. anaphoric relations).

I. Meaning vs. Content.

Another example: Active vs. Passive Sentences;

- (3a) and (4a): different meaning (different anaphora resolution preferences), same content.

(3) a. At 1 AM last night, Veronica was yelling at Josephine.

b. She was drunk again (i.e. Veronica).

(4) a. At 1 AM last night, Josephine was being yelled at by Veronica.

b. She was drunk again (i.e. Josephine).

I. Meaning, Content and Context.

Meaning determines content *given a context.*

(5), when uttered on a Thursday in a discussion about John, entails (6).

(5) **He** came back **three days ago**.

(6) (Therefore) **John** came back **on a Monday**.

We need the utterance context to determine the contents of (5) and (6) and the entailment relation.

I. Meaning, Content and Context.

Meaning changes the context.

(7) a. A man came in.

b. He sat down.

(7a) changes the context relative to which the content of (7b) is determined: the man that came in is the one that sat down.

I. Accessing Content and Determining Entailment in Discourse.

We need to determine entailment relations **within** discourse,

e.g. if we want to give lexical entries for particles like *hence* and *therefore*, which express such discourse internal relations.

(8) a. A man came in.
b. Therefore, he entered.

I. Accessing Content and Determining Entailment in Discourse.

In sum, to determine entailment relations…

- we need to access contents **within** discourse;
- content is context-dependent and the context changes in discourse.

I. Other content- and entailment-based phenomena in natural language.

There are other natural language phenomena indicating that we need explicit access to contents and entailment relations **within** discourse.

I. Questions and Entailment.

Answerhood conditions are content-based.

Entailment ($\models$) determines what counts as an answer:

ϕ is an answer to question $?\psi$ **iff** there is world w such that $\|\phi\|^{M} \models \|?\psi\|^{M,\,w}$

(see Groenendijk & Stokhof (1997), Def. 4.19)

I. Presupposition and Entailment.

Cleft presuppositions are content-based.

(note the active-passive alternation)

(9) A: … was the invitation to New York for which I did not apply. **I was just invited**.

(35 lines later)

A: **It was he who invited me**.

(see Spenader (2000): 2, Example 2)

I. Discourse Relations and Entailment.

Emphasis.

(10) It’s unfortunate that **it's cloudy in San Francisco this week**, but **CLOUDY IT IS** – so we might as well go listen to the LSA papers.
(see Walker (1993): 27, (26))

(11) **I don't like to go down that way**. It may be shorter, but **I DON'T LIKE IT**.
(see Walker (1993): 199, (105))

II. Explicitly Representing Contents in Discourse – three arguments.

Three arguments for explicit content representation in discourse:

- the adjacency requirement on *therefore / hence* discourses;
- content anaphora: *this*, *the former*, the *latter*;
- more complex cases of modal subordination.

II. Explicitly Representing Contents: *therefore / hence* and adjacency.

Therefore is felicitous only if it is adjacent to the premise that entails the conclusion:

(12) John came in. Therefore, John entered.
(13) #John came in. Mary left. Therefore, John entered.

… much like a non-restrictive relative clause needs to be adjacent to its **antecedent**:

(14) John, who was at home, called to say he had the flu.
(15) #John called to say he had the flu, who was at home.

II. Explicitly Representing Contents: content anaphora (*this*, *former*, *latter*)

(16) a. One plaintiff was passed over for promotion three times.

b. Another didn't get a raise for five years.

c. But the jury didn't believe this.

(based on an example in Asher & Lascarides (2003): 15)

The demonstrative *this* in (16c) can refer back to the content of (16b), but not to the content of (16a)…

… which is unexpected if contents are not explicitly represented in discourse.

II. Explicitly Representing Contents: content anaphora (*this*, *former*, *latter*)

The same point is established if we replace *this* in (16c) with *the latter*.

The expressions *the former* and *the latter* "[…]have built into them an explicit instruction to select the leftmost or the rightmost of two potential antecedents. But how are we to describe this in a theory which implies that the selection of an antecedent is not required or even pertinent for the interpretation of the anaphor?"

(Heim (1990): 169)

II. Explicitly Representing Contents: complex cases of modal subordination.

(17) a. You should buy a lottery ticket and put it in a safe place.

b. You're a person with good luck.

c. It might be worth millions.

(example from Roberts (1996): 224)

The modal quantification in (17c) is interpreted as restricted by the content of the first conjunct 'below' the modal *should* in (17a):

(given that you're a generally lucky person…)

if you buy a lottery ticket, it might be worth millions.

(…hence, you should buy one and put it in a safe place)

II. Explicitly Representing Contents: complex cases of modal subordination.

Explicit discourse reference and anaphora to contents seems to be a fairly constrained way to derive the intuitively correct interpretation of (17c) …

… if compared to the arguably unrestricted accommodation of quantifier domain restrictions that Roberts (1996) advocates.

III. Constraints on Content and Meaning Representation in Discourse.

- content is determined relative to the context;

- the context changes in discourse and the meaning is what changes the context;

- the context needs to encode **structured** plural anaphora between sentences;

- there is structured plural anaphora in both the **individual** and the **modal** domain.

III. Content and Meaning Representation: Entailment in Context.

We determine contents with respect to a context: the premise in (5) entails the conclusion in (6)…

… if (5) is uttered on a Thursday in a discussion about John.

(5) **He** came back **three days ago**.

(6) (Therefore) **John** came back **on a Monday**.

III. Content and Meaning Representation: Entailment and Anaphora.

Moreover, the context changes in discourse:

premises change the context with respect to which the content of the conclusion is determined.

(8) a. **A man** came in.

b. (Therefore) **He** entered.

(8a) entails (8b) if the anaphor ***he*** refers to the same thing (i.e. the same discourse referent) as the antecedent ***a man***.

III. Content and Meaning Representation: Entailment and Plural Anaphora.

(18) **Every man** saw **a woman.**

(19) (Therefore) **They** noticed **them**.

(18) entails (19) if the plural anaphors ***they*** and ***them*** refer back to all (salient) men and the women they saw...

… in a structured way.

III. Content and Meaning Representation: Entailment and Plural Anaphora.

(18) **Every man** saw **a woman.**

(19) (Therefore) **They** noticed **them**.

In a structured way, that is:

Entailment does not obtain if man m_1 saw woman n_1 and m_2 saw n_2…

but we interpret (19) as asserting that m_1 noticed n_2 and m_2 noticed n_1.

III. Content and Meaning Representation: Entailment and Plural Anaphora.

(18) **Every man** saw **a woman.**

(19) (Therefore) **They** noticed **them**.

The seeing and the noticing have to be **structured** in the same way.

The **pairing** / **distribution** of individuals in the two sentences has to be the same.

III. Content and Meaning Representation: Entailment and Modal Anaphora.

The anaphoric connections between premises (20a-20b) and conclusion (21) can be **modal**:

(20) a. A wolf **might** come in.
b. It **would** see John.
(21) (Therefore) It **would** notice him.

(20a) introduces a possible scenario: a wolf comes in;
(20b) elaborates on this scenario: said wolf sees John;
(21) is entailed only if it is interpreted with respect to the same possible scenario.

III. Content and Meaning Representation: Entailment and Modal Anaphora.

(20) a. A wolf **might** come in.

b. It **would** see John.

Modal anaphora is **plural**. (20) says that:

- there is an epistemic possibility *p* in which a wolf *u* comes in and sees John;
- **any** epistemic possibility *p* in which a wolf *u* comes in is such that the wolf *u* sees John.

(see Roberts (1989))

III. Content and Meaning Representation: Entailment and Modal Anaphora.

(20) a. A wolf **might** come in.

b. It **would** see John.

Modal anaphora is **structured**.

If we consider possibility p_1 in which wolf u_1 enters and possibility p_2 featuring intruder u_2,

then (20b) asserts that p_1 features the same wolf u_1 (and not u_2!) seeing John (similarly for p_2 and u_2).

III. Content and Meaning Representation: Entailment and Modal Anaphora.

(20) a. A wolf **might** come in.   b. It **would** see John.
(21) (Therefore) It **would** notice him.

Structured plural modal anaphora is relevant for entailment (just as individual-level plural anaphora is).

The seeing in (20b) and the noticing in (21) have to be **structured** in the same way.
The **pairing** / **distribution** of possibilities and wolves has to be the same for entailment to obtain.

III. Content and Meaning Representation: Summary.

To determine **entailment** (answerhood, presupposition etc.) **between two contents**,

i.e. to determine that one content contains at least as much **factual information** as the other, we need:

- to explicitly *store* and *access contents* in discourse;
- to account for the fact that:
  - (a) determining content is *context sensitive* and the context stores *structured pluralities*;
  - (b) the premises *change the context* relative to which we determine the content of the conclusion.

IV. Dynamic Plural Logic with Contents (DPLC).

The central observation:

Modal subordination provides a paradigm for accessing contents in discourse and determining relations between them.

In particular,

entailment relations hold between suitably selected **plural modal discourse referents**,

which are used to store and access contents in a context-sensitive way.

IV. Dynamic Plural Logic with Contents (DPLC).

Building on previous work on modal and plural individual-level anaphora (Stone (1999) and van den Berg (1996)):

I propose a new dynamic system in which both the **static contents** and the **dynamic meanings** of sentences can be represented.

The same formal machinery is used to account for both entailment and modal subordination.

The anaphoric parallels between the individual and the modal domain are explicitly captured.

IV. Dynamic Plural Logic with Contents (DPLC).

The Structure of the Section.

- General features of the system (type logic, information states, discourse referents)
- Why plural information states
- Representing contents and meanings: the **max** (maximizing) operator
- Modal subordination as modal anaphora
- Defining truth and entailment; lexical entries for entailment particles (*therefore* / *hence*)

IV. DPLC: Many-sorted Type Logic.

Following Muskens (1995, 1996), the dynamic system is formulated in **many-sorted type logic**.

Basic types (I ignore the temporal and eventuality domains):

- type *t*: truth-values;
- type *e*: individuals;
- type *s*: models variable assignments;
- type **w**: possible worlds.

IV. DPLC: Info States and Discourse Referents.

- **info states *I*, *J*, *K*,…** are **sets of 'variable assignments'**, i.e. they are of type *st*.

  (following van den Berg (1996))

- an **individual discourse referent** (dref) ***u*** is of type *se* and it stores a **plurality** with respect to an info state ***I***:

  (22) $uI := \{x_e: \exists i_s \in I_{st}\ (x=ui)\}$

  (the subscripts on terms indicate their type)

  Simplifying: $uI := \{ui: i \in I\}$

IV. DPLC: Info States and Discourse Referents.

- a **modal discourse referent** (dref) ***p*** is of type $s\mathbf{w}$ and it stores a **proposition** (a set of worlds) with respect to an info state ***I***:

(23) $pI := \{w_{\mathbf{w}} \colon \exists i_s \in I_{st}\, (w = pi)\}$
(the subscripts on terms indicate their type)

Simplifying: $pI := \{pi \colon i \in I\}$

IV. DPLC: Why Plural Info States?

Why do we model drefs for pluralities and propositions in this way and not via drefs for sets?

Their type would be $s(et)$ for pluralities of individuals and $s(\mathbf{w}t)$ for propositions (contents).

Because we need to capture **structured inter-sentential plural anaphora:**

Plural info states (type st) are able to store and pass on **the internal distributive structure of pluralities.**

IV. DPLC: Why Plural Info States?

(18) Every man saw a woman.

- it asserts that for each and every man, there is a woman that he saw: m_1 saw n_1, m_2 saw n_2 etc.

- so, the output info state after processing (18) looks something like this…

IV. DPLC: Why Plural Info States?

I	u_1 (men)		u_2 (women)	…
i_1	m_1 $(=u_1i_1)$	$\xrightarrow{m_1 \text{ saw } n_1}$	n_1 $(=u_2i_1)$	…
i_2	m_2 $(=u_1i_2)$	$\xrightarrow{m_2 \text{ saw } n_2}$	n_2 $(=u_2i_2)$	…
i_3	m_3 $(=u_1i_3)$	$\xrightarrow{m_3 \text{ saw } n_3}$	n_3 $(=u_2i_3)$	…
…	…	…	…	…

each assignment i_1, i_2, … stores a particular man and a particular woman that stand in the *see*-relation.

IV. DPLC: Why Plural Info States?

I	u_1 (men)		u_2 (women)	…
i_1	m_1 (=u_1i_1)	$\xrightarrow{m_1 \text{ saw } n_1}$	n_1 (=u_2i_1)	…
i_2	m_2 (=u_1i_2)	$\xrightarrow{m_2 \text{ saw } n_2}$	n_2 (=u_2i_2)	…
i_3	m_3 (=u_1i_3)	$\xrightarrow{m_3 \text{ saw } n_3}$	n_3 (=u_2i_3)	…
…	…		…	…

for each $i \in I$, the man in i saw the woman in i.

IV. DPLC: Why Plural Info States?

(18) **Every man** saw **a woman.**

(19) (Therefore) **They** noticed **them**.

Since the plural info state I is passed on to (19), we are able to retrieve the *seeing* structure:

for each $i \in I$, the man in i saw, **hence noticed**, the woman in i (and not some other woman).

IV. DPLC: Representing Contents and Meanings.

Sentence **Contents** (propositions): modal drefs - type $s\mathbf{w}$.

Sentence **Meanings**: Discourse Representation Structures (DRSs),

i.e. relations between info states of type $(st)((st)t)$.

e.g. $[p, u \mid come\_back_p\{u\}]$ is a DRS;

p – modal dref; u – individual dref.

IV. DPLC: Representing Contents and Meanings.

Sentence meanings are **relations** between plural info states and not functions because we non-deterministically assign pluralities to the new plural drefs just as we non-deterministically assign individuals to new singular drefs:

(24) a. A boy entered. b. He was wearing a blue sweater.

(25) a. Some boys entered. b. They were wearing blue sweaters.

(26) a. At least one boy wrote a letter to Santa.
b. He wanted a new bicycle.

(27) a. At least two boys wrote letters to Santa.
b. They wanted new bicycles.

IV. DPLC: Representing Contents and Meanings.

DRSs – their general format is:

[new drefs, e.g. p, u | **conditions**, e.g. $come\_back_p\{u\}$**]**

The 'linearized box' notation abbreviates:

$\lambda I_{st} J_{st}.\ I$[**new drefs**]$J$ & **conditions**J

i.e. the output state ***J*** differs from ***I*** at most with respect to the **new drefs** and each **condition** is satisfied by the output state ***J***.

IV. DPLC: Representing Contents and Meanings.

The actual definitions:

- $i[u_0, \ldots, u_n]j := \forall u\, (\mathbf{dref}(u)\ \&\ u \neq u_0\ \&\ \ldots\ \&\ u \neq u_n \rightarrow ui = uj)$

  where **dref** is a constant with the intuitive interpretation "is a dref".

- $I[u_0, \ldots, u_n]J := \forall j \in J\, \exists i \in I\, (i[u_0, \ldots, u_n]j)\ \&$
  $\forall i \in I\, \exists j \in J\, (i[u_0, \ldots, u_n]j)$

- $[u_0, \ldots, u_n \mid C_0, \ldots, C_m] := \lambda IJ.\ I[u_0, \ldots, u_n]J\ \&$
  $C_0J\ \&\ \ldots\ \&\ C_mJ$

IV. DPLC: Representing Contents and Meanings.

(6) **John** came back **on a Monday**.

Content: $\{w\text{: } come\_back_w(john)\}$

i.e. the *maximal* set of worlds *w* in which John comes back (I ignore the temporal issues)

come_back is an intensional property of type **w**(*et*), i.e. it relates worlds and individuals (atomic or not).

see, for example, is an intensional relation of type **w**(*e*(*et*)).

IV. DPLC: Representing Contents and Meanings.

(6) **John** came back **on a Monday**.

$\{w: come\_back_w(john)\}$

We introduce a ***max*** operator over modal drefs to be able to extract and store this proposition.

Meaning of (6): **max**^{p} ([| $come\_back_p\{John\}$])

Content of (6): encoded by the modal dref p

IV. DPLC: Representing Contents and Meanings.

$[\mid \textit{come\_back}_p\{John\}] := \lambda IJ.\ I[\ ]J\ \&\ \textit{come\_back}_p\{John\}J$

where:

$I[\ ]J$ is just identity, i.e. $I{=}J$

- $\textit{come\_back}_p\{John\} := \lambda I_{st}.\ \forall i{\in}I\ (\textit{come\_back}_{pi}(John\,i))$

- $John := \lambda i_s.\ john$

  ($John$ – dref, type se; $john$ – individual constant, type e)

  (see Muskens (1996) for this analysis of proper names)

IV. DPLC: Representing Contents and Meanings.

I=J	***p*** **(worlds)**	***John***	…
i_1	w_1 $(=pi_1)$	$john$ $(=ui_1)$	…
i_2	w_2 $(=pi_2)$	$john$ $(=ui_2)$	…
…	…	…	…

$\forall i \in I\ (come\_back_{pi}(John i))$, i.e.

$$come\_back_{w_1}(john),\ come\_back_{w_2}(john)\ \ldots$$

IV. DPLC: Representing Contents and Meanings.

$\mathbf{max}^p$ ([| *come_back*$_p${*John*}])

$\mathbf{max}^p$ (D) – 'dynamic λ-abstraction over worlds'

- the 'abstracted variable' is the modal dref *p*
- the 'scope' is the DRS D
- we extract a set of worlds *pJ* (where *J* is the output info state) such that:

  (a) *pJ* contains **only** worlds *w* that 'satisfy' D;

  (b) *pJ* contains **all** the worlds *w* that 'satisfy' D,

  i.e. *pJ* is the *maximal* set of worlds that 'satisfy' D.

IV. DPLC: Representing Contents and Meanings.

$$\mathbf{max}^{p}(\mathrm{D}) := \lambda IJ.\ \exists H\,(\,I[p]H\ \&\ \mathrm{D}HJ\,)\ \&\ \forall K\,(\,\exists H(I[p]H\ \&\ \mathrm{D}HK) \rightarrow pK \subseteq pJ\,)$$

- 1st conjunct: we introduce p as a new dref (symbolized by $I[p]H$) and make sure each world in pJ 'satisfies' D (by $\mathrm{D}HJ$)

- 2nd conjunct: maximality; any other set pK that 'satisfies' D is included in pJ.

IV. DPLC: Representing Contents and Meanings.

$\mathbf{max}^p$ ([| $come\_back_p\{John\}$]) :=

$\lambda IJ.\ I[p]J\ \&\ pJ \subseteq \{w{:}\ come\_back_w(john)\}\ \&$

$\forall K\ (I[p]K\ \&\ pK \subseteq \{w{:}\ come\_back_w(john)\} \rightarrow pK \subseteq pJ\ )$

$\lambda IJ.\ I[p]J\ \&\ pJ = \{w{:}\ come\_back_w(john)\}$

$\mathbf{max}^p$ (D) is always a subset of the DRS [p|]; D,

where ';' is dynamic conjunction, interpreted as relation composition: D; D' := $\lambda IJ.\ \exists H(\mathrm{D}IH\ \&\ \mathrm{D}'HJ)$.

IV. DPLC: Representing Modal Subordination.

All these ingredients are independently needed for representing modal subordination.

(20) a. A wolf **might** come in.

b. It **would** see John.

- there is an epistemically accessible possible world *w* in which a wolf *x* comes in and sees John;
- any epistemically accessible possible world *w* in which a wolf *x* comes in is such that the wolf *x* sees John.

IV. DPLC: Representing Modal Subordination.

(20) a. A wolf **might** come in.

b. It **would** see John.

- we need ***maximality*** (we consider ***any*** epistemically accessible world in which a wolf comes in);

- we need **structured** pluralities: if wolf x_1 enters in world w_1 and x_2 in world w_2, sentence (20b) requires w_1 to be such that x_1 (and not x_2!) sees John.

IV. DPLC: Representing Modal Subordination.

(20) a. A wolf **might** come in.

$\mathbf{max}^{p}$ ([| $p \subseteq p_0$]; [u | $wolf_p\{u\}$, $come\_in_p\{u\}$])

- ***might*** introduces the *maximal* possibility ***p*** in which some wolf ***u*** comes in, i.e. ***p*** collects any world *w* in which there is a wolf that comes in;

- $\boldsymbol{p_0}$ provides the contextually specified set of *epistemically accessible* possible worlds.

IV. DPLC: Representing Modal Subordination.

J	p (worlds)	u (wolves)	…
j_1	w_1 ($=pj_1$)	x_1 ($=uj_1$)	…
j_2	w_2 ($=pj_2$)	x_2 ($=uj_2$)	…
…	…	…	…

- we establish a structured correspondence between worlds and intruding wolves via the conditions:

$$wolf_p\{u\}J := \forall j \in J\ (wolf_{pj}(uj))$$

$$come\_in_p\{u\}J := \forall j \in J\ (come\_in_{pj}(uj))$$

IV. DPLC: Representing Modal Subordination.

(20) b. It **would** see John.

$[\ |\ see_p\{u, John\}]$

- ***it*** is an individual anaphor, referring back to the intruding wolf ***u***; ***would*** is a modal anaphor, referring back to the maximal possibility ***p***
- sentence (20b) simply tests that, in each world *w* in possibility *p*, the corresponding wolf *x* sees John
- Stone (1999) first proposed to analyze modal subordination as anaphora to drefs for **static** modal objects (namely accessibility relations)

IV. DPLC: Representing Modal Subordination.

(20) a. A wolf **might** come in. b. It **would** see John.

$\mathbf{max}^{p}$ ([| $p \subseteq p_0$]; [u | $wolf_p\{u\}$, $come\_in_p\{u\}$]); [| $see_p\{u, John\}$]

- this representation is still too weak to capture the maximizing interpretation of the discourse in (20);

- we need to maximize over both possible worlds and wolves, i.e. for **any** world and **any** wolf in that world that happens to come in, the wolf sees John;

- the representation does not capture the universal quantification over wolves.

IV. DPLC: Representing Modal Subordination.

- we need to maximize over the dref *u* in addition to *p*;
- thus, we maximize over indefinites in both the modal and the individual domains – independently motivated by:

(28) a. Harry bought **some sheep**. b. Bill vaccinated **them**.

(a slightly modified version of example (10) in Evans (1977):104)

Bill vaccinated **all** the sheep that Harry bought and not only some of them – just as in (20) we consider all the epistemic possibilities in which a wolf comes in.

IV. DPLC: Representing Modal Subordination.

(20) a. A wolf **might** come in.

$\mathbf{max}^p$ ([| $p \subseteq p_0$]; ${}_p\mathbf{max}^u$ ([| $wolf_p\{u\}$, $come\_in_p\{u\}$]))

b. It **would** see John.

[| $see_p\{u, John\}$]

${}_p\mathbf{max}^u$ (D) := $\lambda IJ.\ pI{=}pJ\ \&\ \forall w{\in}pI\ (\ \mathbf{max}^u\text{(D)}\, I_{p=w} J_{p=w}\ )$,

where $I_{p=w}$:= $\{i{\in}I : pi{=}w\}$

(for distributivity in the individual domain, see van den Berg (1996) and Nouwen (2003))

IV. DPLC: Representing Modal Subordination.

As far as p_0 (the set of current epistemic possibilities) is concerned, the **might** update is a test:

"[…] sentences of the form ***might***ϕ provide an invitation to perform a test on σ rather than to incorporate some new information in it."

(Veltman (1996): 245)

I assume this analysis for the rest of the presentation, but it is not the only analysis possible in DPLC (see the Appendix).

IV. DPLC: Intermediate Summary.

- we use **modal drefs** to store propositions.

- we use **maximal** modal drefs to store contents and maximal possibilities (see modal subordination).

- we use plural info states to store **structured propositions** / **contents** and **structured sets of individuals**…

…so that we can predict the correct interaction between modal and individual anaphora, i.e. to account for modal subordination and entailment.

IV. DPLC: Defining Truth and Entailment.

TRUTH. A text ***T*** with a meaning of the form $\mathbf{max}^{p}$ (D),
where ***p*** is the designated content referent of ***T***,
is **true** w.r.t. a world ***w*** and an input info state (context) ***I***
iff there is an output state ***J*** s.t. $\mathbf{max}^{p}$ (D)***IJ*** and $\boldsymbol{w} \in \boldsymbol{pJ}$.

i.e. ***pJ*** is the content expressed by ***T*** in context ***I*** and we check that this content (proposition) is true in world ***w***.

IV. DPLC: Defining Truth and Entailment.

e.g. *John came back* is true w.r.t. context I and world w

iff $\mathbf{max}^p$ $([\ |\ come\_back_p\{John\}])$ is true w.r.t. I and w

iff $\exists J$ ($\mathbf{max}^p$ $([\ \ |\ come\_back_p\{John\}])$ IJ & $w \in pJ$)

iff $\exists J$ ($I[p]J$ & $pJ=\{w'\text{: } come\_back_{w'}(john)\}$ & $w \in pJ$)

iff $w \in \{w'\text{: } come\_back_{w'}(john)\}$

iff $come\_back_w(john)$

IV. DPLC: Defining Truth and Entailment.

ENTAILMENT.

A text ***T*** with a meaning of the form $\mathbf{max}^{p}$ (D) **entails** a text ***T'*** with a meaning of the form $\mathbf{max}^{p'}$ (D') w.r.t. an input info state (context) ***I***

iff there is an intermediate state ***H*** and an output state ***J*** s.t. $\mathbf{max}^{p}$ (D)***IH*** and $\mathbf{max}^{p'}$ (D')***HJ*** and $(\boldsymbol{p} \sqsubseteq \boldsymbol{p'})\boldsymbol{J}$.

where $\boldsymbol{p} \sqsubseteq \boldsymbol{p'}$ is **structured inclusion** (a 'closure' requirement):

$$\boldsymbol{p} \sqsubseteq \boldsymbol{p'} := \lambda I_{st}.\ \forall \boldsymbol{i} \in \boldsymbol{I}\ (\ \exists \boldsymbol{i'} \in \boldsymbol{I}\ (\boldsymbol{i}[\boldsymbol{p'}]\boldsymbol{i'}\ \&\ \boldsymbol{p'i'} = \boldsymbol{pi'})\ )$$

IV. DPLC: Defining Truth and Entailment.

Intuitively, ***pJ***(=***pH***) is the content expressed by ***T*** in context ***I***, ***p′J*** is the content expressed by ***T′*** in the intermediate context ***H*** (crucial for the anaphoric connections between premises and conclusion) and the content of ***T*** is at least as informative as the content of ***T′***.

Structured inclusion is necessary for entailment cases involving anaphoric dependencies between premises and conclusion: since these anaphoric dependencies are captured via **structured plural** anaphora, the two contents should also be related in a structured way.

IV. DPLC:
Defining Truth and Entailment.

Entailment in natural language should be defined in terms of **contents** (propositions) and not in terms of **meanings** (context-change potentials) because:

- meaning is 'too fine-grained' for entailment: it contains other kinds of information besides **factual information**;
- whether entailment obtains **depends on the context** – and content, but not meaning, is context-dependent;
- we need to determine entailment relations **within** discourse (within the 'object language', not at the 'meta-language' level).

IV. DPLC: Defining Truth and Entailment.

(29) John came back.

$\mathbf{max}^{p}$ ([| $come\_back_p\{John\}$])

$\lambda IJ.\ I[p]J$ &

$pJ=\{w: come\_back_w(john)\}$

(30) (Hence) John returned.

$\mathbf{max}^{p'}$ ([| $return_p\{John\}$])

$\lambda IJ.\ I[p']J$ &

$p'J=\{w': return_{w'}(john)\}$

(29) entails (30) w.r.t. an input context I

iff $\exists HJ\ (\|(29)\|IH\ \&\ \|(30)\|HJ\ \&\ (p \sqsubseteq p')J\ )$

iff $\exists J\ (\ I[p, p']J\ \&\ pJ=\{w: come\_back_w(john)\}\ \&$
$p'J=\{w': return_{w'}(john)\}\ \&\ (p \sqsubseteq p')J\ )$

IV. DPLC: Defining Truth and Entailment.

(29) entails (30) w.r.t. an input context I

iff $\exists J$ ($I[p, p']J$ & $pJ=\{w: come\_back_w(john)\}$ &
$p'J=\{w': return_{w'}(john)\}$ & $(p \sqsubseteq p')J$)

iff $\exists J$ ($I[p,p']J$ & … & $\forall j \in J$ ($\exists j' \in J$ ($j[p']j'$ & $p'j'=pj'$))

iff $\{w: come\_back_w(john)\} \subseteq \{w': return_{w'}(john)\}$

IV. DPLC: Defining Truth and Entailment.

What guarantees that the meaning representations for the texts ***T*** and ***T'*** are of the form $\mathbf{max}^{p}$ (D) and $\mathbf{max}^{p'}$ (D')?

What guarantees that the DRSs D and D' contain the modal drefs ***p*** and ***p'*** at all and in the suitable configuration for the intuitively correct contents to be stored in them?

I assume that the compositional interpretation of the texts ***T*** and ***T'*,** based on suitable logical forms and lexical entries, yields the correct meaning representations

(cf. the translations from English to Ty_2 (Gallin (1975)))

IV. DPLC: Defining Truth and Entailment.

(8) a. **A man** came in.

$\mathbf{max}^{p}\,({}_{p}\mathbf{max}^{u}\,([\;|\;man_{p}\{u\},\ come\_in_{p}\{u\}]))$

b. (Therefore) **He** entered.

$\mathbf{max}^{p'}\,([\;|\;enter_{p'}\{u\}])$

(8a) entails (8b) w.r.t. an input context I

iff $\exists HJ\,(\|(8a)\|IH\ \&\ \|(8b)\|HJ\ \&\ (p \sqsubseteq p')J\,)$

iff $\exists S_{\mathbf{w}t}P_{\mathbf{w}et}\,(\,S=\{w\!: \exists x_{e}\,(man_{w}(x)\ \&\ come\_in_{w}(x))\}\ \&$

$\forall w\in S\,(Pw=\{x\!: man_{w}(x)\ \&\ come\_in_{w}(x)\})\ \&$

$\forall w\in S\ \forall x\in Pw\,(enter_{w}(x))\,)$

IV. DPLC: Defining Truth and Entailment.

Similarly, we are able to capture the **structured** anaphoric dependencies and the entailment between (18) and (19):

(18) Every man saw a woman. (19) (Therefore) They noticed them.

(*every* is represented as: ${}_{p}\mathbf{max}^{u}$ ([| man_{p}{u}]))

and between (20) and (21):

(20) a. A wolf might come in. b. It would see John.

(21) (Therefore) It would notice him.

IV. DPLC: Lexical entries for *Therefore* / *Hence*.

Lexical entries for *therefore* / *hence* – **two** components:

(a) they are anaphoric to the content of the premises;

(b) they test that the content of the premises is included (in a structured way) in the content of the conclusion.

A text of the form ***Therefore$_p$ T'*** (or *Hence$_p$ T'*),

where ***p*** is the anaphorically retrieved content of the premises,

is interpreted as **max**$^{p'}$ (D'); [| $\boldsymbol{p} \sqsubseteq \boldsymbol{p'}$],

where **max**$^{p'}$ (D') is the meaning of the conclusion ***T'***.

V. Conclusion.

- I have described a dynamic system (DPLC) in which we can explicitly represent the **contents** (propositions) expressed by sentences in discourse and in which we can properly relate static contents, dynamic meanings and discourse context;

- DPLC explicitly encodes the conceptual distinction between meaning and content and analyzes both modal subordination and discourse-internal entailment relations (expressed by *therefore* / *hence*) in terms of contents;

V. Conclusion.

- DPLC follows the research program of Muskens (1995, 1996) and couches the dynamic system in van den Berg (1996) and Nouwen (2003) in many-sorted type logic;

- thus, DPLC makes possible a fairly simple (Montague-style) compositional approach to the interaction between inter-sentential plural anaphora and generalized quantification;

V. Conclusion.

- DPLC extends the dynamic systems in van den Berg (1996) and Nouwen (2003) to the modal domain, following the insights in Stone (1999);

- the extension simplifies Stone (1999) and allows for a straightforward incorporation of the classical approach to modal quantification in Lewis (1973) and Kratzer (1981);

- DPLC explicitly formalizes the parallels between structured plural anaphora and quantification in the individual and modal domains;

V. Conclusion.

- the structured inclusion requirement defined for entailment purposes enables us to capture the intuitive correspondence between 'therefore' discourses and conditional structures;

- DPLC opens the way for an analysis of more complex modal subordination cases involving anaphora to contents (e.g. the example from Roberts (1996)).

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VI. Appendix. The Parallel between '*Therefore*' Discourses and Conditionals.

Intuitively, the entailment relation established in discourse (31) below obtains **iff** the conditional update in (32) is successful.

(31) a. A man came in. b. Therefore, he entered.

(32) If a man came in, he entered.

Arguably, the analysis of (32) is…

VI. Appendix. The Parallel between '*Therefore*' Discourses and Conditionals.

(32) If a man came in, he entered.

$\mathbf{max}^p({}_p\mathbf{max}^u([\ |\ man_p\{u\}, come\_in_p\{u\}]))$; $[\ |\ enter_p\{u\}]$

- the conditional contributes two updates (first proposed in Stone (1999));
- the 1^{st} update ('the restrictor') introduces the content of the antecedent;
- the 2^{nd} update ('the nuclear scope') represents the consequent as anaphoric to the content of the antecedent.

VI. Appendix. The Parallel between '*Therefore*' Discourses and Conditionals.

(32) If a man came in, he entered.

$\mathbf{max}^p({}_p\mathbf{max}^u([\ |\ man_p\{u\},\ come\_in_p\{u\}]))$; $[\ |\ enter_p\{u\}]$

The conditional in (32) is basically analyzed as:

(33) a. **Suppose** a man came in. b. **Then**, he entered.

Then is a content anaphor: it accesses the content of the previous 'supposition'.

VI. Appendix. The Parallel between '*Therefore*' Discourses and Conditionals.

(32) If a man came in, he entered.

$\mathbf{max}^p({}_p\mathbf{max}^u([\mid man_p\{u\}, come\_in_p\{u\}]));\quad [\mid enter_p\{u\}]$

The consequent of the conditional **tests** if the **pointwise** correspondence between *p*-worlds and *u*-men established in the antecedent satisfies the consequent.

Which is exactly what the structured inclusion requirement $p \sqsubseteq p'$ in the definition of entailment ensures.

VI. Appendix. The Parallel between '*Therefore*' Discourses and Conditionals.

For an input state (context) I, entailment obtains,

i.e. there is an intermediate state H and an output state J s.t. $\mathbf{max}^{p}(\mathrm{D})IH$ and $\mathbf{max}^{p'}(\mathrm{D'})HJ$ and $(p \sqsubseteq p')J$,

iff the corresponding conditional update is successful,

i.e. there is an intermediate state H and an output state J s.t. $\mathbf{max}^{p}(\mathrm{D})IH$ and $[\mathrm{D'}]^{p/p'}HJ$,

where $[\mathrm{D'}]^{p/p'}$ is exactly like D' except that we substitute p for each occurrence of p' in D'.

VI. Appendix. The Parallel between Ontological Domains.

DPLC explicitly captures the parallel between different ontological domains,

following the research program in Partee (1973), Partee (1984), Stone (1999), Bittner (2001), Schlenker (2005) among others,

and extends it to **structured plural anaphora**.

p_0

VI. Appendix. The Parallel between Ontological Domains.

The parallel between plural cross-sentential anaphora in the individual and the modal domains is reflected in:

- types: individual drefs have type *se*, modal drefs have type *s***w**;
- the use of plural info states to account for structured plural anaphora in both the individual and the modal domain;
- the use of the ***max*** operator to account for the maximality of both individual-level and modal plural anaphora (**some sheep** – **them**, **might** – **would**);

p_0

VI. Appendix. The Parallel between Ontological Domains.

- the same kind of tripartite structure is used in both individual-level and modal quantification.

(33) Every man left.

$_{p^*}\mathbf{max}^u$ ([| $man_{p^*}\{u\}$]); [| $leave_{p^*}\{u\}$],

where p^* is the designated dref for the actual world.

(34) If it is raining, John is singing in the rain.

$\mathbf{max}^p$ ([$|p \subseteq p_0$]; [| $raining_p$]); [| $sing\_in\_rain_p\{John\}$],

where the Context Set p_0 restricts the indicative antecedent

p_0

VI. Appendix. The Parallel between Ontological Domains.

In both cases, the quantification has a tripartite structure:

- the universal quantificational force, given by the ***max***^{u} and ***max***^{p} operators;
- the restrictor, i.e. the DRSs in the scope of the ***max*** operators;
- the nuclear scope, i.e. the DRSs sequenced after the ***max*** updates and which are anaphoric to the drefs contributed by the ***max*** updates.

p_0

VI. Appendix. The Parallel between Ontological Domains.

The tripartite structure of the quantification in both the modal and the individual domains is captured by the scheme:

$\mathbf{det}^{u}$(D); D'

where u is an individual or modal dref, D is the restrictor and D' is the nuclear scope that is anaphoric to the dref u.

p_0

VI. Appendix. The Parallel between Ontological Domains.

The general scheme for defining $\mathbf{det}^{\upsilon}$(D) (using **max**!):

$$\mathbf{det}^{\upsilon}(\mathrm{D}) := \lambda IJ.\ \exists H\ (\ I[\upsilon]H\ \&\ \mathrm{D}HJ\ )\ \&$$
$$\exists K\ (\ \mathbf{max}^{\upsilon}(\mathrm{D})\ IK\ \&\ \mathbf{DET}(\upsilon K,\ \upsilon J)\ )$$

where **DET** is the corresponding static determiner.

e.g. *most* in the individual domain and the *generic operator* in the modal domain can be obtained if **DET** has its usual definition $\mathbf{MOST}(X, Y)$ **iff** $|X\cap Y|>|X\backslash Y|$

VI. Appendix. Representing Conditionals.

(34) If it is raining, John is singing in the rain.

$\mathbf{max}^{p}$ $([\ |p \subseteq p_0]; [\ |\ raining_p]); [\ |\ sing\_in\_rain_p\{John\}]$

This representation of conditionals makes them **tests** with respect to the **context set** $\boldsymbol{p_0}$ (just like ***might*** $\boldsymbol{\phi}$ sentences): in each information state, we do **not** narrow down the set of p_0 worlds stored in that state, we simply test whether the p_0-set satisfies the conditional.

However, across information states, we **do** narrow down the **context set**: we rule out all the info states in which the p_0-set does not satisfy the conditional.

VI. Appendix. Representing Conditionals.

But this is not the only possible analysis in the system…

Heim (1983) – the CCP of ***if*** is:

$\mathbf{c}$ + **if** A, B = $\mathbf{c}$ \ ($\mathbf{c}$ + A \ $\mathbf{c}$ + A + B)

(34) If it is raining, John is singing in the rain.

$\mathbf{max}^{p}$ ([$|p \subseteq p_0$]; [| $raining_p$]);

$\mathbf{max}^{p'}$ ([$|p' \subseteq p$]; [| $sing\_in\_rain_p\{John\}$]); $p_0 = p_0 \setminus (p \setminus p')$,

where $p_0 = p_0 \setminus (p \setminus p')$:= $\lambda IJ.\ J = \{i \in I\text{: } p_0 i \notin (pI \setminus p'I)\}$

VI. Appendix. Representing Conditionals.

OR: we can analyze conditionals as in Lewis (1973)/Kratzer (1981), i.e. as tripartite modal quantificational structures restricted by an ordering source. This analysis:

- can assign to a conditional a meaning of the form $\mathbf{max}^{p}(D)$, hence a content;
- therefore, conditionals are not tests: they have a content with which we can intersectively narrow down the current context set;
- the analysis generalizes to non-epistemic (i.e. deontic or circumstantial) modal quantification.

VI. Appendix. Representing Conditionals.

Generalized Limit Assumption:

$$\forall w\, \forall S\, \forall{<}\, (\, \forall w'{\in} S\, (\, \exists w''{\in} S\, (\neg\exists w'''{\in} S(w'''{<_w}w'')\ \&\ (w''{<_w}w' \vee w''{=}w'))\,)\,),$$

where $w''{<_w}w' := {<}(w)(w'')(w')$, i.e. for any world w, ${<}(w)$ is a strict partial order and ${<}(w)(w'')$ is the set of worlds 'strictly less ideal' than w''.

Given the **Generalized Limit Assumption**, the following **Ideal** function is well-defined:

For any w, S and $<$: $\mathbf{Ideal}_w^{<}(S) := \{w' : \neg\exists w''{\in} S(w''{<_w}w')\}$

VI. Appendix. Representing Conditionals.

(35) If there is a murder, the murderer must go to jail.

(36) $\mathbf{Ideal}_{w^*\!<'}(\{w\text{: } \exists x\ (murder_w(x))\}) \subseteq$

$\{w\text{: } \forall y(\ murderer_w(y, x) \rightarrow go\_to\_jail_w(y)\ )\}$,

where w^* is the actual world and <' is the salient deontic OS.

[this static representation fails to account for the 'donkey' anaphora]

(37) ${}_{p^*}\mathbf{max}^{p}({}_{p}\mathbf{max}^{u}([\ |\ murder_p\{u\}]))$; $[p'|\ \mathbf{nec}_{p^*}{}^{<'}(p, p')]$;

${}_{p',u}\mathbf{max}^{u'}([\ |\ murderer_p\{u', u\}])$; $[\ |\ go\_to\_jail_{p'}\{u'\}]$,

where p^* is the dref for the actual world.

VI. Appendix. Representing Conditionals.

(38) If a wolf entered the cabin yesterday, it must have left by now.

(39) $\mathbf{Ideal}_{w^*}^{<}(\mathbf{CS}\cap\{w\text{: } \exists x\ (wolf_w(x)\ \&\ enter\_yst_w(x))\})$

$\subseteq \{w\text{: } leave\_by\_now_w(x)\}$,

where w^* is the actual world, **CS** is the current context set and < is the salient epistemic OS.

[this static representation fails to account for the 'donkey' anaphora]

(40) ${}_{p^*}\mathbf{max}^{p}([\ |p\sqsubseteq p_0];\ {}_{p}\mathbf{max}^{u}([\ |\ wolf_p\{u\},\ enter\_yst_p\{u\}]));$

$[p'|\ \mathbf{nec}_{p^*}^{<'}(p,\ p')];\ [\ |\ leave\_by\_now_{p'}\{u\}]$

where p^* is the dref for the actual world and p_0 is the dref for the CS.

VI. Appendix. Representing Conditionals.

To be able to define the above conditions and updates, we need to extend our inventory with the impossible world ☼ that Stalnaker (1968) introduces (for technical convenience) in his modal account of conditionals.

- the impossible world ☼ is the *exception* object meant to encode partial functions in a system that uses only total functions; following van den Berg (1996), the exception object in the domain D_e of individuals is ⋆;
- # will be used as a meta-variable over exception objects of any sort.

VI. Appendix. Representing Conditionals.

We also need to redefine structured inclusion as follows:

$u \sqsubseteq u' := \lambda I.\ \forall i \in I\ [(ui = u'i \vee ui = \#)\ \&\ (u'i \in (uI - \{\#\}) \rightarrow u'i = ui)]$

(cf. definition **2.4** in van den Berg (1996): 131)

Modal relations are defined as:

$\mathbf{nec}_{p^*}^{<}(p, p') := \lambda I.\ \forall w \in p^*I\ [(p' \sqsubseteq p)\, I_{p^*=w}\ \&\ p'I_{p^*=w} = \mathbf{Ideal}_w^{<}(pI_{p^*=w})]$

$\mathbf{pos}_{p^*}^{<}(p, p') := \lambda I.\ \forall w \in p^*I\ [(p' \sqsubseteq p)\, I_{p^*=w}\ \&\ p'I_{p^*=w} \subseteq \mathbf{Ideal}_w^{<}(pI_{p^*=w})]$

The above notion of structured inclusion $\sqsubseteq$ is too strong for entailment purposes; we redefine entailment using $\Subset$:

$p \Subset p' := \lambda I.\ \forall i \in I\ (pi = p'i \vee pi = \#)$

VI. Appendix. Representing Conditionals.

I leave open the issue of how to formally relate:

- the notion of context set used in the first two analyses of conditionals, i.e. CS as the set updated / **determined** by the content of sentences and containing the plausible candidates for the actual world – see Stalnaker (1978);

and

- the notion of context set used in the Lewis/Kratzer analysis of conditionals, i.e. CS as restricting the domain of modal quantification, hence as the set **determining** the content of certain modal sentences – see Stalnaker (1968, 1975).

VI. Appendix. Representing Conditionals.

(17) a. You should buy a lottery ticket and put it in a safe place. (b. You're a person with good luck.)

c. It might be worth millions. (from Roberts (1996): 224)

To be able to analyze this example, we need to define conjunction as:

- introducing the content drefs of its two conjuncts

and

- requiring a contextually salient modal dref to be a structured subset of each of the two content drefs.

VI. Appendix. Representing Conditionals.

(41) John came in and Mary left.

(42) $\mathbf{max}^{p}([\,|\,come\_in_{p}\{John\}];\ \mathbf{max}^{p'}([\,|\,leave_{p}\{Mary\}]);$

$[\,|\,p^{*}\sqsubseteq p,\ p^{*}\sqsubseteq p']$,

where p^{*} is the dref for the actual world.

(17) a. You should buy a lottery ticket and put it in a safe place.

(43) ${}_{p^{*}}\mathbf{max}^{p'}([u|\ lottery\_ticket_{p}\{u\},\ buy_{p}\{Hearer,\ u\}]);$

${}_{p^{*}}\mathbf{max}^{p''}[\,|\,put\_in\_safe\_place_{p''}\{Hearer,\ u\}];$

$[p|\ \mathbf{nec}_{p^{*}}{}^{<'}(p_{0},\ p)];\ [\,|\,p\sqsubseteq p',\ p\sqsubseteq p'']$,

where p^{*} is the dref for the actual world, p_{0} is the dref for the Context Set and <' is the salient deontic OS.

VI. Appendix. Representing Conditionals.

(17) c. It might be worth millions.

(44) $[p'''|\ \mathbf{pos}_{p^*}{}^{<}(p', p''')]$; $[\ |\ \mathit{worth\_millions}_{p'''}\{u\}]$,

where p^* is the dref for the actual world and $<$ is the salient epistemic OS.

VI. Appendix: The Formal System (DPLC)

This is the formal system without the extensions and modifications needed for the incorporation of the Lewis (1973) - Kratzer (1981) analysis of modal quantification.

- we use sets of 'assignments' as info states and not simply 'assignments';
- the type of 'assignments': *s*; variables: *i*, *j*, *k*…;
- the type of info states: *st*; variables: *I*, *J*, *K*…;
- we use u, u', u'', …, u_0, u_1, u_2, … for individual-level drefs (type *se*), *p*, p', p'', …, p_0, p_1, p_2, … for modal drefs (type *s***w**).

VI. Appendix: The Formal System (DPLC).

The set of basic static types **BasSTyp**:

- $\{t, e, \mathbf{w}\}$ (truth-values, individuals and possible worlds).

The set of static types **STyp**:

- the smallest set including **BasSTyp** and such that, if $\sigma,\tau\in$ **STyp**, then $(\sigma\tau)\in$ **STyp**.

The set of discourse referent types **DRefTyp**:

- the smallest set such that, if $\sigma\in$ **STyp**, then $(s\sigma)\in$ **DRefTyp**.

The set of basic types **BasTyp**:

- **BasSTyp**$\cup\{s\}$ ('variable assignments').

The set of types **Typ**:

- the smallest set including **BasTyp** and such that, if $\sigma,\tau\in$ **Typ**, then $(\sigma\tau)\in$ **Typ**

VI. Appendix: The Formal System (DPLC).

Axioms (based on Muskens (1995, 1996)):

1. dref(u),

for any dref name u of any type $\tau \in$ **DRefTyp.**

2. **dref**(u) & **dref**(u') $\rightarrow u \neq u'$,

for any two different dref names u and u' of any type $\tau \in$ **DRefTyp**.

3. (Identity of 'assignments'): $\forall i_s j_s\ (\ i[\ ]j \rightarrow i=j\ )$

4. ('Enough States' Axiom)

$\forall i_s\ \forall u_{s\tau}\ \forall f_\tau$ (**dref**(u) $\rightarrow \exists j_s\ (i[u]j\ \&\ uj=f)$),

for any dref name u of any type $(s\tau) \in$ **DRefTyp**.

VI. Appendix: The Formal System (DPLC).

1. Abbreviation:

$i[u_0, \ldots, u_n]j := \forall u\,(\mathbf{dref}(u)\ \&\ u \neq u_0\ \&\ \ldots\ \&\ u \neq u_n \rightarrow ui = uj)$

2. Abbreviation:

$I[u_0, \ldots, u_n]J := \forall j \in J\ \exists i \in I\ (i[u_0, \ldots, u_n]j)\ \&$

$\forall i \in I\ \ \exists j \in J\ (i[u_0, \ldots, u_n]j)$

3. DRS definition (a relation between info states – type (*st*)((*st*)*t*)):

$[u_0, \ldots, u_n \mid C_0, \ldots, C_m] := \lambda IJ.\ I[u_0, \ldots, u_n]J\ \&\ C_0 J\ \&\ \ldots\ \&\ C_m J,$

where $C_0, \ldots, C_m$ are conditions of type $(st)t$.

4. DRS sequencing:

$D;D' := \lambda IJ.\ \exists H\ (\ DIH\ \&\ D'HJ\ )$

5. Abbreviation:

$uI := \{ui\!: i \in I\}$

VI. Appendix: The Formal System (DPLC).

6. Abbreviation:

$I_{u=a} := \{i \in I\text{: } ui = a\}$

7. Lexical relations.

$R_p\{u_0, \ldots, u_n\} := \lambda I_{st}.\ \forall i \in I\ (\ R_{pi}(u_0 i, \ldots, u_n i)\ )$

$sing\{u\} := \lambda I_{st}.\ \forall i,i' \in I\ (ui = ui')$

$sing_p\{u\} := \lambda I_{st}.\ \forall i,i' \in I\ (pi = pi' \rightarrow ui = ui')$

$sing_{p,u}\{u'\} := \lambda I_{st}.\ \forall i,i' \in I\ (pi = pi'\ \&\ ui = ui' \rightarrow u'i = u'i')$

8. Maximization.

$\mathbf{max}^{u}(\mathrm{D}) := \lambda IJ.\ \exists H(I[u]H\ \&\ \mathrm{D}HJ)\ \&\ \forall K\ (\ \exists H(I[u]H\ \&\ \mathrm{D}HK) \rightarrow uK \subseteq uJ\ )$

${}_{u}\mathbf{max}^{u'}(\mathrm{D}) := \lambda IJ.\ uI = uJ\ \&\ \forall a \in uI\ (\ \mathbf{max}^{u'}(\mathrm{D})\, I_{u=a} J_{u=a}\ )$

$\mathbf{max}(\mathrm{D}) := \lambda IJ.\ \mathrm{D}IJ\ \&\ \forall K\ (\mathrm{D}IK \rightarrow K \subseteq J)$

VI. Appendix: The Formal System (DPLC).

9. Generalized determiners.

$\mathbf{det}^{\upsilon}(\mathrm{D}) := \lambda IJ.\ \exists H\ (\ I[\upsilon]H\ \&\ \mathrm{D}HJ\ )\ \&\ \exists K\ (\ \mathbf{max}^{\upsilon}(\mathrm{D})IK\ \&\ \mathbf{DET}(\upsilon K,\ \upsilon J)\ ),$

where **DET** is the corresponding static determiner.

$_{\upsilon}\mathbf{det}^{\upsilon'}(\mathrm{D}) := \lambda IJ.\ \upsilon I{=}\upsilon J\ \&\ \forall a{\in}\ \upsilon I\ (\ \mathbf{det}^{\upsilon'}(\mathrm{D})I_{\upsilon=a}J_{\upsilon=a})$

$\mathbf{det}(\mathrm{D}) := \lambda IJ.\ \mathrm{D}IJ\ \&\ \exists K\ (\ \mathbf{max}(\mathrm{D})IK\ \&\ \mathbf{DET}(K,\ J)\ ),$

where **DET** is the corresponding static determiner.

10. Structured inclusion.

$\upsilon{\sqsubseteq}\upsilon' := \lambda I_{st}.\ \forall i{\in}\ I\ (\ \exists i'{\in}\ I\ (i[\upsilon']i'\ \&\ \upsilon'i'{=}\upsilon i')\ )$